

No.	Unit	Case Study Topic	Case Scenario / Organization	Key Concepts Covered	I4C Requirement	IP India Requirement	ISO/IEC 27001: 2022 Requirement	Student Work / Analysis	Mandatory Tables / Diagrams	Final Output
1	Unit 1	Cyber Risk Assessment and ISO 27001 Security Framework for a Data-Driven Organization	Organization manages customer, employee and financial data	Information assets, CIA Triad, threats, vulnerabilities, risks, attack surface	Identify relevant cyber threats and incident-reporting considerations	Identify important organizational digital assets/IP where applicable	ISMS, information-security risk assessment, security controls	Identify assets → threats → vulnerabilities → risks → CIA impact → controls	Asset Inventory, Risk Matrix, CIA Matrix, ISO Mapping, Security Framework	Complete risk assessment + proposed ISO 27001-based security framework
2	Unit 1	Phishing and Identity Theft Incident Analysis: Detection, Response and Prevention	Employees receive fraudulent emails and credentials are compromised	Social engineering, phishing, identity theft, authentication, authorization	Identify cybercrime type, reporting mechanism and evidence	Identify misuse/copying of organizational digital content if relevant	Access control, authentication, awareness, incident management	Analyse phishing lifecycle → compromised identity → impact → response → prevention	Attack Flow, Evidence Table, CIA Analysis, Risk Matrix, Control Mapping	Phishing Incident Response & Prevention Plan
3	Unit 1	Ransomware Attack and Cybercrime Reporting: Incident Response Case Study	Organization loses access to business files because of ransomware	Threats, vulnerabilities, data breach, backup, recovery, risk	Identify ransomware category, reporting process, evidence and immediate response	Identify affected digital assets/IP where applicable	Incident management, backup, recovery, risk treatment and business continuity	Analyse attack → affected assets → evidence → recovery → preventive controls	Attack Lifecycle, Risk Matrix, Incident Response Flow, Backup Plan	Ransomware Incident Response Framework
4	Unit 1	MFA, Password Security and Access Control for Organizational Data Protection	Unauthorized access occurs because of weak passwords	Authentication, authorization, passwords, MFA, identity protection	Identify relevant cybercrime scenario and evidence	Identify protection of software/digital assets where relevant	Access control, authentication information, identity management, awareness	Compare existing authentication with proposed MFA/RBAC solution	Access-Control Matrix, Risk Matrix, Authentication Flow	Secure Authentication & Access Control Framework
5	Unit 2	Personal Data Protection Across the Data Lifecycle	E-commerce organization collects customer personal and payment-related information	Personal data, PII, sensitive data, collection, storage, processing, sharing, disposal	Identify possible cybercrime/privacy-related incidents	Identify ownership/protection of digital assets where relevant	Information classification, data protection, access control, secure handling	Trace data from collection → storage → processing → sharing → disposal	Data Lifecycle Diagram, Classification Table, Risk Matrix	Secure Data Lifecycle Framework
6	Unit 2	Data Breach and Privacy Risk Analysis in Social Media and Digital Analytics	Digital platform exposes personal/user analytics data	Digital footprint, social-media privacy, privacy risks, data analytics	Identify online/social-media cybercrime and reporting considerations	Identify copyright/trade mark/design issues associated with digital content where applicable	Risk assessment, access control, information protection, incident management	Identify privacy risks → breach impact → response → prevention	Privacy Risk Matrix, Data Flow, CIA Analysis, Control Mapping	Digital Privacy & Data Protection Plan

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7	Unit 2	Protection of Software and Digital Assets Using Intellectual Property Rights	Startup discovers unauthorized copying of software/source code/logo/content	Data classification, information assets, copyright, patent, trademark, design	Identify cybercrime/security aspects of unauthorized access or theft	Identify applicable Copyright / Patent / Trademark / Design protection using official IP India resources	Asset protection, access control, information-security risk management	Identify digital assets → ownership → IPR → security risks → protection mechanisms	Digital Asset Register, IPR Mapping, Risk Matrix, Protection Framework	Digital Asset & IPR Protection Framework
8	Unit 3	Cloud Security and Shared Responsibility Model: Security Assessment	Organization moves customer applications/data to cloud	Cloud computing, shared responsibility, access control, cloud risks	Identify relevant cloud/cybercrime threats and incident response	Identify ownership/protection of software and digital assets where relevant	Risk assessment, access control, cloud-related security controls, supplier/security management	Separate provider/customer responsibilities → identify gaps → propose controls	Shared Responsibility Matrix, Risk Matrix, Cloud Security Architecture	Cloud Security Assessment & Control Framework
9	Unit 3	AI-Assisted Threat Detection and Security of AI Systems	Organization uses AI for threat detection but faces poisoning/adversarial risks	AI cybersecurity, threat detection, automation, data poisoning, adversarial attacks	Identify cyber threats and incident-reporting relevance	Identify ownership/IP considerations for AI software, datasets or generated assets where applicable	Risk management, access control, information protection, security monitoring	Analyse AI security architecture → threats → model/data risks → controls	AI Security Architecture, Threat Matrix, Risk Matrix, Control Mapping	AI Security & Threat Detection Framework
10	Unit 3	Generative AI, Deepfakes and Emerging Cybersecurity Risks	AI-generated content is used for impersonation, fraud or unauthorized digital content	Deepfakes, GenAI risks, impersonation, emerging threats	Identify applicable cybercrime category, evidence and reporting process	Analyse copyright/trademark/design considerations for digital content	Risk assessment, incident management, access control, information protection	Analyse incident → evidence → risk → I4C response → IP considerations → controls	Incident Flow, Risk Matrix, I4C Mapping, IP Mapping, ISO Mapping	Generative AI & Deepfake Security Framework

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Common	All Units	Required Case Study Methodology	Each group works on one assigned case	Information security + privacy + cybercrime + IPR + ISMS	Use official I4C/National Cyber Crime Reporting Portal information	Use official IP India information	Use ISO/IEC 27001:2022 concepts and controls	1. Case Background → 2. Problem Identification → 3. Asset Identification → 4. Threat Identification → 5. Vulnerability Identification → 6. Risk Assessment → 7. CIA Analysis → 8. I4C Analysis → 9. IP India Analysis → 10. ISO 27001 Mapping → 11. Corrective Action → 12. Preventive Action → 13. Final Framework → 14. Conclusion	Asset Table + Threat/Vulnerability Table + Risk Matrix + CIA Matrix + I4C Mapping + IP India Mapping + ISO Mapping + Framework Diagram	15–20 page Case Study Report + 8–10 Slide Presentation + Viva
Common	All Units	Mandatory Risk Assessment	Applied to every case	Threat, vulnerability, likelihood, impact, risk treatment	Include cybercrime-related risk where applicable	Include IP-related risk where applicable	Map identified risks to relevant ISO 27001 requirements/controls	Risk Score = Likelihood × Impact	Risk Register	Risk Treatment Plan
Common	All Units	Mandatory I4C Investigation	Applied according to the case	Cybercrime identification and response	Identify crime type, reporting route, evidence, immediate response, prevention	—	Map incident-management requirements	Identify relevant I4C resources without conducting a real cyberattack	I4C Analysis Table + Incident Response Flow	I4C-Based Cyber Incident Response
Common	Units 2–3 mainly	Mandatory IP India Investigation	Particularly important for software, digital content and IP cases	Copyright, patent, trademark, design	Identify cybercrime aspect if IP is stolen/misused	Identify relevant IPR and official IP India information	Protect information assets and control access	Identify asset → IPR → ownership → risk → protection	IP Asset Register + IPR Mapping Table	IP Protection Plan
Common	All Units	Mandatory ISO 27001 Mapping	Organization-level ISMS perspective	Risk management, CIA, controls, security processes	—	—	Map identified risks to relevant ISO/IEC 27001:2022 requirements/controls	Identify gap → proposed security control → expected security improvement	ISO Mapping Table	ISO 27001-Based Security Framework

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Common	All Units	Final Integrated Framework	Combine findings from the entire case	CIA + Risk + I4C + IP India + ISO 27001	Cybercrime response	IPR protection	ISMS/security controls	Assets → Threats → Vulnerabilities → Risk → CIA → I4C → IP India → ISO 27001 → Corrective Action → Preventive Action	One complete framework/architecture diagram	Final Organizational Information & Data Security Framework
Common	All Units	Submission Requirements	Group-based case study	Research, analysis, documentation, presentation	Official I4C source references	Official IP India source references	Official ISO source/reference + academic sources	Report must contain evidence-based analysis rather than only theory	15–20 pages + 8–10 slides	Report + Presentation + Viva